| II SEMESTER | L | T | P | C |
|---|---|---|---|---|
| | 3 | - | - | 3 |
| **20IT2T01: IT ESSENTIALS** | | | | |

**COURSE OUTCOMES:**

**After the completion of this course, students will be able to**

**CO1:** Understand the concepts of operating systems.
**CO2:** Demonstrate the implementation of various software engineering tools.
**CO3:** Understand the basics of Internet.
**CO4:** Understand the orientation towards web basics.
**CO5:** Demonstrate the implementation of various computer graphics concepts.

**UNIT-1**

**Operating Systems:** Introduction, Functions of an operating System, Classification of Operating Systems, System programs, Application programs, Utilities, Installation of Operating Systems, The Unix Operating System, Basic Unix commands.

**UNIT-2**

**Software Engineering:** The evolving role of software, changing nature of software, software myths, Structure of Software Life Cycle, Software engineering methodologies, software requirements, various software engineering tools.

**UNIT-3**

**Internet Basics:** Introduction, Features of Internet, Internet applications, Services of Internet, Logical and physical addresses, Internet Service Providers, Domain Name System, Security- Forms of attacks, legal approaches to network security.

**UNIT-4**

**Web Basics:** Introduction to web, web browsers, web servers, Protocol, HTTP/HTTPS, TCP/IP, Email, FTP, SMTP, SNMP, URL, HTML and CSS.

**UNIT-5**

**Computer Graphics:** Scope of Computer Graphics, Overview of 3D Graphics, Modelling- modelling individual objects, modelling entire scenes. Dealing with global lighting - Ray tracing, Radiosity. Rendering and Animation.

**TEXT BOOKS**

1. J. Glenn Brookshear, “Computer Science: An Overview”, Addision-Wesley, TwelfthEdition, 2014.
2. Pradeep K Simha, “Computer Fundamentals- Concepts, Systems & Applications”, 8th edition, BPB.